

Ananya Bajaj

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EDUCATION

University of California, San Diego

Sep. 2024 - Jun. 2028

Bachelors of Science, Data Science

Coursework: Data Structures/Algorithms, Theoretical Fundamentals of Data Science, Linear Algebra, Statistics for Data Science

Activities: Alpha Kappa Psi Professional Business Fraternity, Data Science Student Society

PROFESSIONAL EXPERIENCE

Data Analyst Intern @ Colibri Digital Marketing

Jun. 2023 - Aug. 2023

- Analyzed and processed datasets using Python (NumPy, pandas) and spreadsheets, improving report accuracy by 20% and delivering actionable insights for local and international clients.
 - Applied data cleaning and aggregation to extract meaningful patterns and trends, strengthen reporting accuracy, and support strategic, data-driven decision-making processes.
 - Authored an article on the benefits of Google Analytics in organizations, effectively translating complex analytical results into clear, actionable recommendations for non-technical stakeholders.
 - Collaborated closely with the analytics team, quickly learning new tools, methodologies, and project workflows to adapt efficiently to diverse assignments and tight deadlines.
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PROJECTS

House Price Prediction using Gradient Descent Algorithm

Dec. 2025

- Built gradient descent algorithm from scratch in Python to train a linear regression model.
- Derived the MSE loss function and gradients to iteratively update model parameters.
- Evaluated model performance using mean squared error and root mean squared error, and analyzed the impact of learning rate and the number of iterations on convergence behavior.

AI-Powered Travel Insight Generator

Dec. 2025

- Built a modular ML pipeline in Python leveraging DSPy and the Wikipedia API for structured sequential data processing, with robust error handling and scalable parallel processing.
 - Implemented a modular, multi-step LLM orchestration pipeline leveraging custom DSPy signatures and parallel processing to enable scalable, efficient, and reliable content generation.
 - Improved output reliability and RAG accuracy through intelligent context composition and robust handling of API failures and Wikipedia disambiguation scenarios.
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TECHNICAL SKILLS

- Python · Java · NumPy · Pandas · SQL · R · Machine Learning · Data Management Cloud · Google Cloud · Git · Data Visualization
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CERTIFICATIONS

- **Cloud Digital Leader Certification**, Google
- **Prepare Data Exploration**, Google
- **Ask Questions to Make Data-Driven Decisions**, Google
- **Foundations: Data, Data, Everywhere**, Google